Exercise 7.9: Given any Alice observable **A** and Bob observable **B**, show that for a product state, the correlation $C(A, B)$ is zero.

From this exercise, we can learn something about entanglement. If a system is in a state where one can find any two observables **A** and **B** that are correlated—meaning that $C(A, B) \neq 0$—then the state is entangled. Correlations are defined to lie in the range −1 to +1. These extreme values represent the greatest possible negative and positive correlations. The greater the magnitude of $C(A, B)$, the more entangled is the state. If $C(A, B) = 0$, then there is no correlation (and no entanglement) at all.

7.7.2 The Density Matrix Test for Entanglement

To calculate correlations, you have to know about both Bob's part and Alice's part of the system, along with the system wave function. But there is another test for entanglement that only requires us to know Alice's (or Bob's) density matrix. Let's suppose that the state $|\Psi\rangle$ is a product state of a Bob factor $|\varphi\rangle$ and an Alice factor $|\psi\}$. That means the composite wave function is also the product of a Bob factor and an Alice factor:

$$\psi(a, b) = \psi(a)\phi(b).$$

Now, let's work out Alice's density matrix. We use the definition in Eq. 7.20 to get

$$\rho_{a'a} = \psi^*(a)\psi(a') \sum_b \phi^*(b)\phi(b).$$

But if Bob's state is normalized, then